DATA ANALYTICS — RESEARCH ASSIGNMENT 2865001730

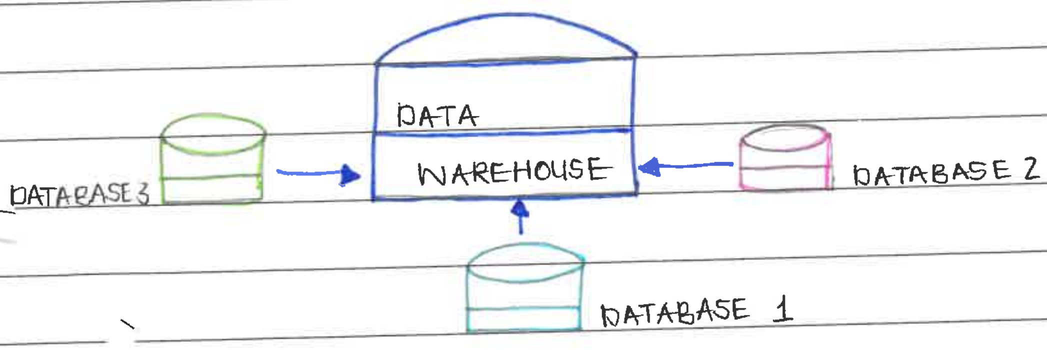
1. A Database is where we store data. It is a place where data is collected, ~~organized~~ so that it can be easily accessed. A Database is essential because it gives you a single and reliable place to store and manage information so that people or systems can use it to retrieve information.

EXAMPLES:

① Healthcare systems — Hospitals store patient data, medical records and appointments. This helps them to track client information and manage the information.

② Social Media — An app like Facebook stores user profiles, pictures and interactions of the app users.

2. A Datawarehouse is a central place designed to store multiple databases



Unlike a regular database, a data warehouse is optimized for analytical queries (OLAP) online analytical processing across large volumes of data.
A datawarehouse is used when you need reliable ~~ces~~ consistent analytics across the business and it can store historical data.

3. A data Lake is a big and scalable storage area that holds raw data without having to "clean up" the data. A Datawarehouse stores structured data whilst a DataLake you can store the data the figure it out later.

Data Lake is usually preferred when storing huge volumes of many different types of data.

4. A datamart is a specialized data storage system focused on a specific data set. For example, a particular business unit (Finance, marketing)
Datamart can be dependant on datawarehouse or not they can or sometimes stored in a datawarehouse. It enables you to get specific insights faster thus typical users are departmental BI teams and Business analysts.

5. Structured data is data with a fixed and organized format mostly as tables with rows and columns.
Example: Spreadsheet with employee records.

Employer ID	Name	Dept

Unstructured data is information that doesn't follow a defined format.
Example: PDF and videos.
Semi-structured data is data with some organized properties. It doesn't follow a fixed format but it still contains markers and indicators.
Example: Email have structured headers and an unstructured body text.

6. Data modeling is the process of defining how data is structured, stored, retrieval and analysis. It is important because it serves as a blueprint for organized data and it ensures data consistency and guides system designs.

① Logical data model — defines data structures & relationships

② ~~Blueprint~~ Conceptual data model — high level, abstract view of data.

7. Data mining is the process of discovering or searching large amounts of data using patterns, trends and connection between information. It uses mostly statistics and machine learning.

Businesses use data mining to discover trends in order to create/make an informed decision.

① Customer Behavior & Sales

Used to get insight on product recommendations and targeted promotions to drive sales

② Forecasting.

A company can mine data ~~using~~ to forecast ~~ing~~ & predict future trends.

8. A Data management system manages retrieve, access, update transactions, store data in a database.

1. MySQL → It is an open source suited for web applications especially in common web stacks

② Oracle Database → It is a commercial source used for large critical systems in finance.

③ MongoDB → Data stored in a document form suitable for flexible formats.

④ PostgreSQL → This is an advanced open source suited for complex queries and advanced features.

9. An ETL is a process of data integration that involves extracting data from different sources then transforming it into a usable format thereafter loading it into a datawarehouse or a specific data base for analysing and report.

This process enables organizations to analyze and report their data effectively and it ensures data quality.

Extract → Gathering data (raw) from various sources

Transform → Extracted data is cleaned, formatted

Load → The clean data loaded to a specific database or datawarehouse.

10. CSV (Comma - Separated Values):

Ideal for simple data, plain text, easy to load

It is primarily used for exporting spreadsheets or databases and is easily readable but inefficient for large scale analytics

JSON (JavaScript Object Notation)

Suitable for structured data specifically in web applicatio

It supports hierarchical data structures and is highly readable by humans and machines making it usable for configuration files.

Parquet

A columnar format that is particulary useful for largescale analytics and data processing, use for big data analytics workloads.

AVRO

A columnar format with scheme support for big data applications.

It is particularly useful for handling large datasets.

11. Artificial Intelligence refers to computer systems or machines capable of performing tasks that ordinary require human intelligence.

General AI possess broad, human-like cognitive abilities across tasks. (A hypothetical system that learn & reason like humans)

Narrow AI is designed for specific tasks (virtual assistance)

12. Machine Learning is a subset of AI where systems learn patterns from data to make predictions or decisions instead of relying on coded rules done by developers.

Traditional learning excels in stable, well defined environments with structured data it uses a hardcoded list of banned words, requiring manuel updates everytime new ~~patterns~~ ~~emerge~~.

TYPES:

1. Supervised: Learns from labeled inputs to predit labels.
2. Unsupervised: This finds patterns or structure in unlabeled data.
3. Reinforcement: Learns by trial and error to maximize rewards in an environment.

13. Deep Learning is a branch of machine learning that uses deep neutral networks to stimulate the decision making processes of the human brain.
Neural networks consist of layers of interconnected neurons with weight; data propagates forward that produces outputs.
Deep learning is one of the most advanced methods within AI. it enables systems to perform tasks that require humanlike understanding. It is also a further specialisation within machine learning using neutral networks to automatically learn features from data.

14. NLP enables computers to understand, interpret and generate human language. This is done by closing the gap between human communication and machine learning.
① Search engines interpreting queries
② Email spam filtering and categorization
③ Chatbots and virtual assistants.

15. LLM is a deep learning model usually based on the transformer architecture, trained on massive amounts of texts to predict the next word and generated human like language.
LLM differ from standard machine learning models by scale and by being pretained on broad unsupervised objectives then cleaned for tasks.
① Open AI models
② Google PaLM

16. Generative AI produces new content such as texts, images, audios by learning the distribution of training data and using it as sample data.

① High-level : Used to train a model to use model data

② Tools such as Chat GPT.

③ Github Copilot.

17. A training dataset is the collection of labeled or unlabeled examples that Machine learning models learns from training. The quality of this data matters a lot. If it ~~must be~~ accurate, representative and diverse then the model more likely to perform well and behave fairly.

If training data is biased or incorrect, the model can learn and amplify those biases and produce unfair outcomes, or make serious mistakes.
Higher errors rates for underrepresented groups or unsafe choices in high stakes systems

18. Computer vision allows machines to see by changing images into numerical representations and then using models to recognize patterns so that the system can detect, classify or track objects in what it sees.

① Automated quality inspection

② Automomous driving perception.

19. An application programming interface is a set of rules and tools that lets software systems communicate.
An API is like a restuarant menu and waiter i.e. you request a menu and a specific dish the waiter delivers your order to the kitchen and returns the results.
APIs enables modularity and integration between services and systems.

20. An API Key is a secret credential that is used to authenticate and identify a client application to an API. It is used to control access, track usage and help avoid unauthorized requests. API must be stored securely and replace/revoke them if there is a chance it was exposed.

21. REST (Representational State Transfer) is an architectural style for web services using stateless communication and standard HTTP methods.
① GET Retrieve a resource.
② POST Create a resource or submit data.
③ PUT Replace or update a resource.
④ DELETE Remove a resource

22. Javascript Object Notation is a lightweight text format for representing structured data as key value pairs and arrays. It is widely useds for APIs because its human-readable

23. Prompt engineering is the skill of writing clear, well structured instructions for an AI model so that it produces the results you actually need.
It is important because the model will follow the of detail you give it.

Example (1)
Poor: summarize ~~this~~ text
Improved: Summarize the following paragraph into 3 points focusing on the main findings

Example (2)
Poor: write code.
Improved: Write a python function that accepts a list of integers and return a new list with duplicates removed while preserving the original order.

24. A map function applies a given function to each element in a collection and returns a list of results.
Example: [Python]
nums = [1,2,3]
squares = list (map)(lamda x:x* nums)) # 1,4,9
Map is used for concise element wise transformations.

25. Cloud computing means using computer services such servers, storage, software over the internet whenever you need them, instead of buying your own equipment.
There are 3 main models:

(1) Infrastructure as a Service
This provides the basic building blocks such as virtual servers, storage and networking.
Example: Amazon.
(2) Platform as a service: managed platform to build and deploy app Example: Google app engine.
(3) Software as a service: ready to use applications delivered over the web
Example: Microsoft 365

26. A version control system tracks changes to files over time.
Git is a distributed VCS widely used by developers and analysts to manage code, collaborate and maintain history.
A commit is a saved snapshot of changes with a message
A branch is an independent line of development allowing parallel work without affecting the main codebase.

27. A Jupyter Notebook functions like a digital workbook where you can write code, add notes and show results all in the same place. You can run the code step by step languages
It is popular because it makes it easy to
(1) Explore and test ideas while analysing data
(2) Create work that can be repeated and checked again
(3) Display charts and results directly next to the code
(4) Explain your work clearly in a step by step way as if one is telling a story.

28 Python is a high level programming language that is easy to read and write. It is interpreted language, this means you can run and test your code without needing to compile it first.

1. NumPy: numerical computing and arrays
2. Pandas: data manipulation and tabular data structures
3. Sckit-learn: traditional ML algorithms and utilities
4. TensorFlow: deep learning frameworks for building neutral networks

29. Data privacy means protecting peoples personal information and making sure their data is handled in a safe and secure way. Respecting their rights over how it is collected, used and shared.

GDPR is a European union data protection law. It applies to any organisation that handles person data of people in the EU/EEA and it gives people rights like accessing their data

30. Data ethics addresses moral issues in data collection, analysis, and usage.

Examples:

1. Using person data without consent for profiling
2. Deploying biased hiring algorithms that discriminate against protection groups

31 Data governance is the set of people, rules and processes an organisation uses to make sure data is owned, understood, trusted, and used safely and correctly across the business

① Data stewardship.
② Data quality management
③ Metadata and cataloging.
④ Access controls.

32 Bias in AI occurs when a model systematically favors or harms certain groups because of skewed or unrepresentative training data, labels or designs choices.
Example: Facial recognition.

33. Data Analyst → Focuses on reporting, dashboards, descriptive analytics; skills SQL, Excel, visualization, tools PowerBI, Tableau, Excel.

Data Scientists → Builds predictive models and experiments
skills: statistics, ML, Pytho/R
tools: scikit-learn, Python, Jupyter.

Data Engineer → Builds data pipelines and Infrastructure
skills: ETL, databases, distributed systems.
tools: Spark, airflow, SQL, cloud storage

AI Engineer → productionizes ML/AI models at scale
skills: ML engineering, model serving, MLOps
tools: TensorFlow / PyTorch, Kubernetes

34. Business intelligence transforms data into actionable insights through dashboards, reports and visual analytics to drive decisions

Tools:

1. Power BI: Microsoft tool for interactive dashboards and enterprise reporting.

2. Tableau: Data visualization tool for exploratory analysis and storytelling. Both allows users to connect to data sources.

35. I found the difference between data lakes and data warehouse most interesting because it highlights different architectural choices — flexibility and structure.

36. A map function = google and python documentation. https://docs.python.org/3/library/functions.

37. AI can automate routine reporting, surface client risk signals from claims and market data. In my current role it will help with freeing time for strategic client work and improving responsiveness

REFERENCES

1. IBM (2023) https://www.ibm.com/topics/database.
2. IBM (2022) https://www.ibm.com/docs/en/data-modeling.
3. Oracle (2024) Oracle Database Documentation https://docs.oracle.com/en/database/
4. SAS (2020) https://www.sas.com/en_us/insights/analytics/data.